

# Types of data: categorical, numerical, ordinal

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In this video, you will learn

- The different interpretations of “data types.”
- The five categories of data types often distinguished in statistics.
- Examples of each type of data.

# But first, a disambiguation

**“Data type” in  
computer science**

Format of data storage

Implications: operations,  
storage space

Examples: floating-point,  
boolean, string

**“Data type” in  
statistics**

Category of data

Implications: Appropriate  
statistical procedures

Examples: next slides!

# Data types distinguished in this course

| Category                | Type     | Description                             | Example                       |
|-------------------------|----------|-----------------------------------------|-------------------------------|
| Numerical<br>(numbers)  | Interval | Numeric scale with meaningful intervals | Temp. in C                    |
|                         | Ratio    | interval, but also with meaningful zero | Height in cm                  |
|                         | Discrete | No arbitrary precision (integers)       | Population                    |
| Categorical<br>(labels) | Ordinal  | Sortable, discrete                      | Education (high school, uni.) |
|                         | Nominal  | Non-sortable, discrete                  | Movie genre (sci-fi, romcom)  |

## More examples

**Discrete** Number of widgets purchased in January 2019.

**Nominal** Last three countries visited.

**Ordinal** Favorite three countries (Vietnam, Fiji, New Zealand).

**Nominal** Gender (male, female, other)

**Ratio** Land area of forests

**Ordinal** Cities sorted by population (Beijing, NYC, Villa Las Estrellas)

**Interval** IQ (intelligence quotient)

**Ordinal** Satisfaction with this course so far (rating 1-5)

## Why does this matter?

Some statistical analyses can be applied only to certain data types.

Some visualization methods are valid only for certain data types.



18: Code: representing types of data on computers

```
22 chardata = 'xyz';
23
24 % string
25 strdata = "x";
26
27 % boolean (aka logical)
28 logitdata = true;
29
30 % cell
31 celldata = { [3 4 34] 'hello' 4 };
```

Command Window

```
>> whos
```

| Name       | Size | Bytes | Class   | Attributes |
|------------|------|-------|---------|------------|
| celldata   | 1x3  | 378   | cell    |            |
| chardata   | 1x3  | 6     | char    |            |
| logitdata  | 1x1  | 1     | logical |            |
| numdata    | 1x4  | 32    | double  |            |
| strdata    | 1x1  | 150   | string  |            |
| structdata | 1x1  | 572   | struct  |            |

1.5x 4:26 / 8:57

stats\_whatRdata\_types

localhost:8889/notebooks/Dropbox%20(Personal)/classes/udemy/statsML/whatRdata/stats\_whatRdata\_types.ipynb

stats\_whatRdata\_types Last Checkpoint: 05/03/2020 (autosaved)

Python 3

File Edit View Insert Cell Kernel Widgets Help

Run

Code

```
In [ ]: ## create variables of different types (classes)

# data numerical (here as a List)
numdata = [ 1, 7, 17, 17 ]

# character / string
chardata = 'xyz'

# double-quotes also fine
strdata = "x"

# boolean (aka logical)
logitdata = True # notice capitalizatin!

# a list can be used like a MATLAB cell
listdata = [ [3, 4, 34] , 'hello' , 4 ]

# dict (kindof similar to MATLAB structure)
dictdata = dict()
dictdata['name'] = 'Mike'
dictdata['age'] = 25
dictdata['occupation'] = 'Nerdoscientist'
```

```
In [ ]: # Let's see what the workspace Looks Like
%who
```

odemy



```
# dict (kindof similar to MATLAB structure)
dictdata = dict()
dictdata['name'] = 'Mike'
dictdata['age'] = 25
dictdata['occupation'] = 'Nerdscientist'
```

```
In [2]: # Let's see what the workspace looks like
%whos
```

| Variable  | Type | Data/Info |
|-----------|------|-----------|
| chardata  | str  | xyz       |
| dictdata  | dict | n=3       |
| listdata  | list | n=3       |
| logitdata | bool | True      |
| numdata   | list | n=4       |
| strdata   | str  | x         |

```
In [3]: # clear the Python workspace
%reset -sf
```



# Sample vs. population data

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In this video, you will learn

- The difference between sample and population data.
- Examples of the two.
- Why this distinction is important.

# Sample vs. population data: definitions

**Population data** Data from ALL members of a group.

**Sample data** Data from SOME members of a group  
(hopefully randomly selected).

# Sample vs. population data: examples

**Population data** Salaries in a department; Number of lions in a zoo; Facebook group members.

**Sample data** Average height of Italians; Salaries of teachers; Brightness of stars in Milky Way.

# Sample vs. population data: examples

Sample data

The number of pets that most Dutch people have.

Population data

The number of Udemy courses completed by sophomores at the University of X.

Population data

The number of people who voted in Lichtenstein in 2020.

Sample data

The opinion of Chinese people about the mountains in Grenoble.

# Sample vs. population notation

Population *parameters*      $\mu, \beta, \sigma^2$

Sample *estimates*      $\hat{\mu}, \hat{\beta}, \hat{\sigma}^2$   
                                  $\bar{x}, \beta, s^2$



# Sample vs. population data: who cares?!

**Most statistical procedures are designed either for sample or for population data.**

Applying a procedure to the wrong data type can lead to incorrect results and incorrect interpretations!

(Most data you'll work with in practice are sample data.)

# Samples, case reports, and anecdotes

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In this video, you will learn

- The differences between  $N=1$  and  $N>1$  studies.
- Why  $N=1$  studies must be interpreted with great caution.

# Sample sizes and names

**$N > 1$**

- Pilot study
- Proof-of-principle
- Small-scale study
- Large-scale study

**$N = 1$**

- Case study: One patient
- Anecdote: One person

# Sample sizes and names

**$N > 1$**

Actual research with potentially important and generalizable findings.

**$N = 1$**

Interesting stories that may inspire future research, but that (most often) should not be overly trusted or generalized.

# Problems with sample sizes

**$N > 1$**

Can suffer from sampling variability, noise, and outliers. Appropriate statistical analyses can facilitate interpretation.

**$N = 1$**

Highly likely to be noise, outliers, or other non-representative data (otherwise why report it?). Statistical inference is difficult or impossible.